

Lab 4: Bias Detective Investigation

Project: TalentMatch Rwanda AI Bias Analysis

Role: QA Engineer, TalentMatch Rwanda

Status: Optional Lab

1. Executive Summary

Bias Severity: High

Primary Affected Groups:

- Female candidates
- Candidates who studied at public/TVET institutions outside the main private-university track (e.g. IPRC campuses, public teacher-training colleges) rather than at University of Rwanda's flagship campuses or well-known private universities
- Candidates from Northern, Western, and Southern Province addresses, relative to Kigali City
- Candidates whose CVs were submitted primarily in Kinyarwanda rather than English or French

Top 3 Recommended Actions:

- Remove or sharply down-weight proxy features (university brand, home province, CV language) that correlate with historical hiring rather than with candidate ability
- Retrain the scoring model on a rebalanced, representative dataset that does not simply relearn the employer base's past hiring pattern
- Stand up weekly fairness monitoring across gender, province, and CV-language groups, not only gender

2. Detailed Findings

2.1 Bias Type Analysis

1. Historical Bias

Present? Yes

Evidence:

- The training data is drawn from five years of actual hiring outcomes across TalentMatch Rwanda's employer clients, which were themselves concentrated in Kigali-based tech and BPO firms — **69% male hires vs 31% female hires** across the training set.
- Candidates whose most recent employer or internship was in Kigali City are overrepresented relative to Rwanda's actual working-age population distribution across provinces.

Disadvantaged Groups:

- Female candidates
- Candidates whose prior work experience was outside Kigali City

2. Sampling Bias

Present? Yes

Evidence:

- The training dataset is heavily skewed toward **software/IT roles (58%)**, with customer-service, sales, and non-tech roles underrepresented even though TalentMatch Rwanda's client base has since diversified into those sectors.
- **74%** of training records have a Kigali City address, even though Kigali holds roughly 15% of Rwanda's population — the model has, in effect, barely seen a rural applicant.

Underrepresented Groups:

- Applicants for non-tech roles
- Applicants from Northern, Western, Southern, and Eastern Province addresses

3. Measurement Bias

Present? Yes

Evidence:

- The model uses "professional network size" (LinkedIn-style connection count) and "reference count" as scoring inputs. Both track access to a professional network built through Kigali-based internships and employer contacts far more than they track a candidate's actual competence — a highly capable graduate from Musanze or Huye with no Kigali internship simply has fewer connections to show, for reasons that have nothing to do with skill.
- CV completeness score penalizes CVs written primarily in Kinyarwanda with English section headers only, treating them as "incomplete" relative to fully bilingual/English CVs — this measures a candidate's exposure to English-medium CV-writing coaching, not their job competence.

Unfair Measurement:

- Professional-network and reference-count signals structurally favor candidates who already had access to Kigali's formal job market before they even applied.

4. Proxy Bias

Present? Yes

Proxy Features:

- University name/brand (flagship private universities and University of Rwanda's Kigali-based colleges score higher than IPRC/TVET or other public campuses, independent of the specific

program studied)

- Home province / address on file
- Professional network size
- CV language mix (English/French-forward CVs score higher than Kinyarwanda-forward CVs)

Impact:

- Indirect discrimination against candidates from lower-income households and rural provinces, who are statistically more likely to have attended TVET/public institutions, to have Kinyarwanda-forward CVs, and to lack a Kigali-built professional network — none of which reflects their actual ability to do the job.

2.2 Bias Pipeline Mapping

[Historical Hiring Decisions by Kigali-concentrated employer clients] ↓ [Training Data Collection] → Bias Point #1: skewed hiring data (gender, province, role type) ↓ [Feature Selection] → Bias Point #2: inclusion of proxy features (university brand, province, network size, CV language) ↓ [Model Training] → Bias Point #3: model learns and amplifies the historical Kigali/male/private-university pattern ↓ [Deployment & Scoring] → Bias Point #4: discriminatory ranking of otherwise-qualified rural, female, and TVET-educated candidates ↓ [Biased Hiring Outcomes at Client Companies]

2.3 Feature Risk Analysis

High-Risk Features:

- University brand/name
- Home province / address
- Professional network size
- CV language mix

Moderate Risk:

- Reference count
- Extracurricular/volunteer activity descriptions

Low Risk:

- Verified technical skills assessment score
- Years of relevant work experience

3. Mitigation Plan

3.1 Immediate Actions (This Week)

Feature Removal:

- Remove or reduce the weight of:
- University brand/name (retain only verified degree level and field of study)
- Home province/address
- Professional network size
- CV language mix / "completeness" scoring tied to English fluency

Threshold Adjustments:

- Introduce fairness-aware scoring thresholds so no single demographic, province, or language group is systematically pushed below the interview-invite line at a materially different rate than others.

Output Monitoring:

Track weekly:

- Gender selection ratio
- Province-of-residence distribution among shortlisted candidates
- CV-language distribution among shortlisted candidates vs. applicant pool

3.2 Short-Term Actions (1–3 Months)**Data Collection:**

- Actively recruit training examples from employer clients outside Kigali City and from non-tech roles, so the model stops learning "Kigali tech hire" as its implicit default profile.

Model Retraining:

- Retrain on the rebalanced dataset with fairness constraints applied during training, not only checked afterward.

Human Oversight:

- Require a human recruiter to review any batch of shortlisted candidates before it is sent to a client, specifically checking for province/gender/language skew.

3.3 Long-Term Actions (6–12 Months)**Fairness Metrics to Adopt:**

- Equal Opportunity (recommended) — are equally-qualified candidates from different groups shortlisted at similar rates?
- Demographic Parity (secondary, monitoring only) — raw shortlist-rate comparison across groups, useful as an early warning even though it does not account for underlying qualification differences.

Process Changes:

- Cap how much weight any single automated score can carry in a final hiring recommendation; require a documented reason whenever the model's top-ranked candidate is not the one referred to the client.

Transparency Measures:

- Disclose to candidates that an AI ranking tool is used in the process.
- Provide candidates a plain-language explanation of the factors the score is based on.
- Offer a channel for candidates to request human review of their ranking.

4. Success Metrics

- Shortlist rate gap between gender groups narrows over successive quarters.
- Shortlist rate for non-Kigali provinces moves toward parity with Kigali City, adjusted for applicant volume.
- CV-language mix of shortlisted candidates approaches the CV-language mix of the overall applicant pool.
- Weekly fairness reports show a consistent, not one-off, improvement trend.

5. Timeline

Phase	Timeline	Key Actions
Immediate	Week 1	Remove/down-weight proxy features, stand up weekly monitoring
Short-Term	1–3 Months	Rebalance training data, retrain with fairness constraints, add human review gate
Long-Term	6–12 Months	Adopt Equal Opportunity metric formally, add candidate-facing transparency measures

6. Conclusion

TalentMatch Rwanda's scoring model demonstrates significant bias risk, and the mechanism is a familiar one: a model trained on a Kigali-concentrated, male-skewed hiring history will reproduce that pattern unless something in the pipeline actively corrects for it. What makes this case distinct from a generic bias audit is the specific shape the proxies take in the Rwandan labor market — home province standing in for urban/rural income differences, and CV language standing in for access to

English-medium schooling — both of which are easy to miss if a fairness review only checks for gender bias and stops there. Removing the highest-risk proxy features, rebalancing the training data to include the non-Kigali, non-tech, and Kinyarwanda-forward candidates the model has barely seen, and adopting Equal Opportunity as the standing fairness metric are the three changes most likely to close the gap this investigation found — and, just as importantly, to keep it from reopening the next time the model is retrained on another year of the same skewed hiring history.